

Econ 2 - Lecture 14 - 5/18/26

Lecture Quiz #7 Released Wednesday

↳ Due Wednesday (!), 5/27

Memorial Day on Monday, 5/25

Discussion Activity #4: Gamification → Thoughts
Signaling in Education? → Passcode on Canvas

Today: Finish Debt (Chapter 5.2), Monetary Policy (Chap 6)

Last Class:

Total Debt = Public Debt + Intragovernmental Holdings
Fact #1

Paying Back Debt → Roll over indefinitely
Fact #2

* Debt rises from 100 to 120

* Interest Rate falls from 6% to 4%

* Nominal GDP Rises from 200 to 240

Min. Tax Rate = $\frac{\text{Interest Rate} \times \text{Debt}}{\text{Nom. GDP}} \times 100$

Year 1 = $\frac{6\% \times 100}{200} \times 100 = \frac{6}{200} \times 100 = 3\%$

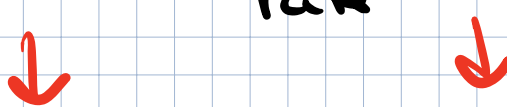
Year 2 = $\frac{4\% \times 120}{240} \times 100 = \frac{4.8}{240} \times 100 = 2\%$

↓
decreasing

Fact #3: Relevant debt statistic = $\frac{\text{Debt}}{\text{GDP}}$ ratio

$$\text{Min. Tax Rate} = \text{Interest Rate} \times \frac{\text{Debt}}{\text{GDP}} \times 100$$

As $\frac{\text{debt}}{\text{GDP}} \uparrow \Rightarrow \text{min. tax rate} \uparrow$



$\frac{\text{Debt}}{\text{GDP}}$ clock is slow, but representative

Fact #4: Interest Payments on Debt
are part of government budget,
not part of government expenditures
 $\Rightarrow$ not part of fiscal policy

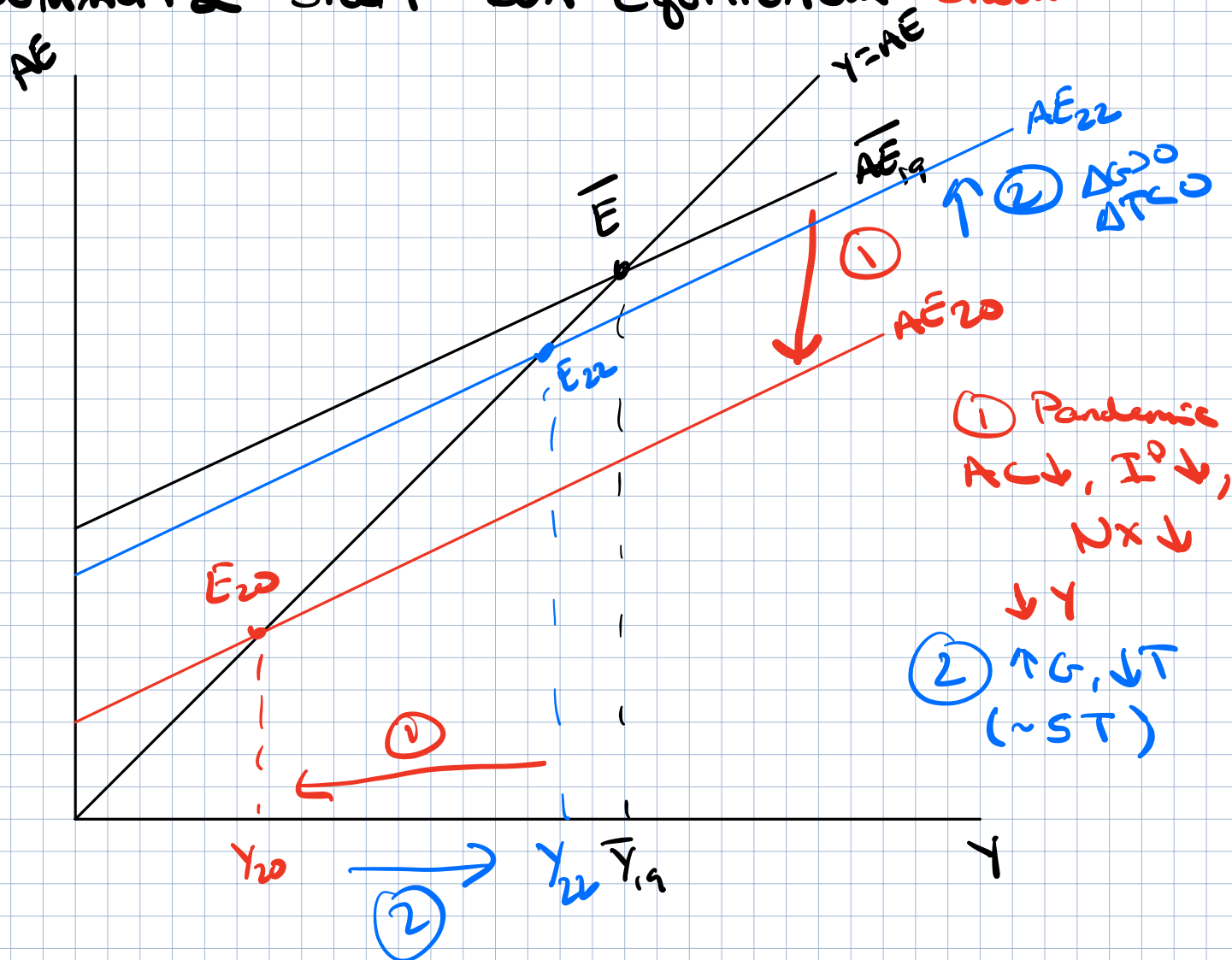
Make a interest payment $\rightarrow$ investor $\Rightarrow$ mpc? Low

$\sim 1/3$ of interest payments $\rightarrow$ foreign holders

Safety / Stability is US Dollar

$\rightarrow$ No requirement that interest is spent anywhere

Summarize: Short-Run Equilibrium + Shock



Between 2019 & 2022, Y was largely unchanged
 Big increase in Debt

$\Rightarrow \frac{\text{Debt}}{\text{GDP}}$ ratio increased!

$$\text{Minimum Tax Rate} = \text{Interest Rate} \times \frac{\text{Sum}(G-T)}{\text{Prices} \times \text{Real GDP}} \times 100$$

How can we improve/reduce the tax burden?

Returned to $\bar{Y}$, but tax rate is higher!

~~1.)~~ Decrease Deficit $\rightarrow \downarrow G \rightarrow \downarrow AE \rightarrow \downarrow Y$
 $\rightarrow \uparrow T \rightarrow \downarrow AE \rightarrow \downarrow Y$

2.) Change the interest rate

$\downarrow$ $\rightarrow$ Reduce interest $\rightarrow$ Reduce min. tax rate
not $\rightarrow$ Monetary Policy ^{rate}
fiscal $\rightarrow$ Higher Prices
policy

~~3.)~~ Increase in prices

Double prices $\rightarrow$ Double the "income"

$\rightarrow$ Associated with reduction in Y

4.) Encourage / incentive increase in Y

If Y increases $\rightarrow$ tech boom $\rightarrow$ internet

If $Y \uparrow$, min. tax rate $\downarrow$

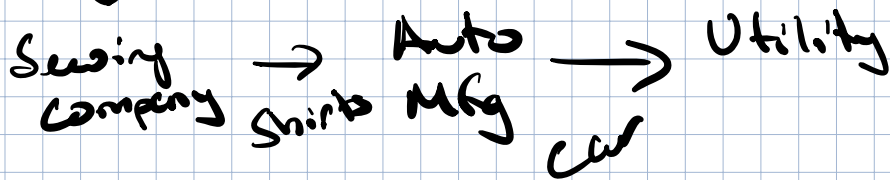
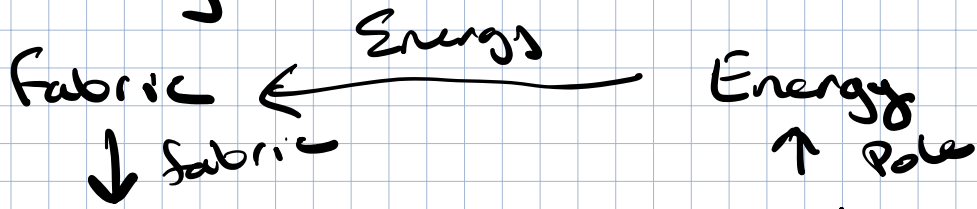
Next Part of Course: Monetary Policy

Same goal as before: Get Y back to $\bar{Y}$

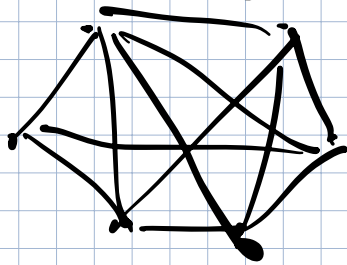
Connect monetary policy actions to $Y = AE$

Define, model, enact, consequences

Money? 1990s → Russia → Bartering system

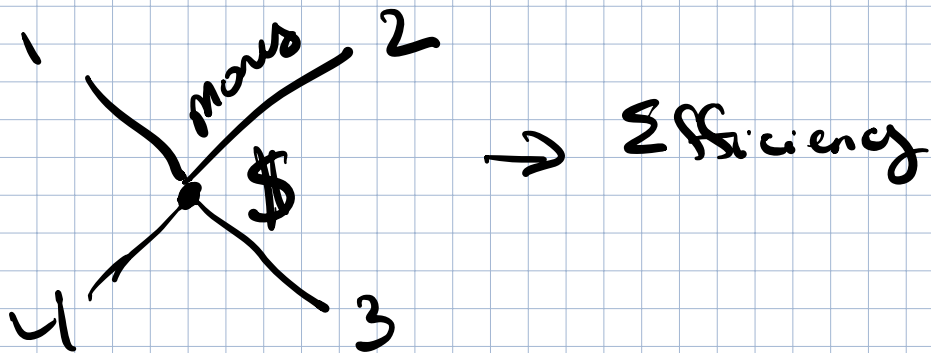


6 goods → $\frac{6 \times 5}{2} = 15$ trading pairs



50 goods
 $\frac{50 \times 49}{2} = 1,225$ pairs

One good that was part of all trades → money



→ Efficiency

3 Main Characteristics of Money

1. Medium of Exchange: buyers & sellers accept
2. Unit of Account: every good is valued by "money"
3. Store of Value: long lasting, preserve value

Examples

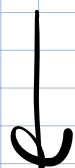
Commodity Money: Money with intrinsic value

Examples of Commodity:

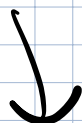
- Cigarettes (specific times, geographics)
- Rai Stones on Yap Island
 - ↳ Mine granite in 1700s, bring back to island
 - ↳ Limit on "stone" inflation
 - ↳ Europeans hyperinflated stones
- Salt trade → Gold, Silver → Rice
- Spices → Silk → Shells, beads

Gold → only mine 1-2% more per year of existing stock

↳ Stable currency



Fiat Money: US Dollar ⇒ no intrinsic value of US Dollar



→ Backed by belief in gov't & usability

Norm today

↳ New concept → Before 1970s, \$35 ⇒ 1 oz of Gold

Bitcoin money?

Digital Version of Rai Stones

Store of Value? Not recently
Unit of Acct? Not now.

Medium of Exchange? Not today